

Mohammadreza Qaderi

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Professional Summary

Senior Backend Engineer and Technical Lead with 5+ years of experience designing high-performance, distributed systems using Go and Python. Proven expertise in microservices architecture, FinTech trading platforms, and engineering scalable EdTech data pipelines capable of processing millions of daily events. Adept at leading cross-functional teams, optimizing cloud-native infrastructure, and integrating AI-driven solutions to drive product vision and robust business outcomes.

Education

Amirkabir University of Technology

Tehran, Iran

B.Sc. in Computer Software Engineering; GPA: 3.89/4.0

Sept 2018 – Jun 2023

Thesis: A Paper Trading System for Iranian Stocks (Developed and deployed in collaboration with GradientDp!)

National Organization for Development of Exceptional Talents (SAMPAD)

Tehran, Iran

Diploma in Mathematics and Physics; GPA: 4.0/4.0

Sept 2014 – Aug 2018

Honors & Awards

- **Top 0.1% Nationwide:** Ranked within the top 0.1% among approximately 170,000 participants in the National University Entrance Exam (Konkour).
- **National Olympiad Finalist (34th Period):** Advanced to the 2nd round of the Iranian National Mathematics and Computer Olympiad.
- **National Olympiad Finalist (33rd Period):** Advanced to the 2nd round of the Iranian National Mathematics and Computer Olympiad.

Professional Experience

Backend Tech Lead & Software Engineer

Aug 2023 – Present

Pishro Broker

Tehran, Iran

- **Microservices Migration & Optimization:** Spearheaded the architectural transition of the core trading platform (my.pishrobroker.ir) from a legacy monolith to a high-performance microservices architecture. Migrated entirely to gRPC to drastically reduce API latency and accelerate data fetching speeds.
- **Financial Services Development:** Architected highly specialized microservices, including the 'Marketer' service for VIP clients' complex financial computations, and the 'Credit' service for managing tiered margin trading allocations.
- **Options Trading Platform (optionz.ir):** Led backend engineering for the options trading system. Engineered real-time data ingestion pipelines from diverse financial databases to perform complex on-the-fly trading calculations.
- **API Gateway & Infrastructure:** Refactored the public-facing API Gateway (gateway.pishrobroker.ir), improving system reliability by integrating comprehensive logging and rigorous testing frameworks.
- **Deployment & CI/CD:** Dockerized the entire ecosystem and established automated CI/CD pipelines utilizing GitLab Runner.
- **Tech Stack:** Python, gRPC, Docker, GitLab CI/CD, RabbitMQ, Redis, PostgreSQL (Frontend: D3.js, TradingView).

Co-Founder & Chief Technology Officer (CTO)

Sefro Yek Farda

Jan 2021 – Present

Tehran, Iran

- **Technical Leadership & Product Vision:** Co-founded the EdTech startup and led the technical strategy for scalable B2B/B2C platforms, successfully supporting over 80,000 active users and educational institutes.
- **AI-Driven Recommendation Engine (entekhab.yejooredigeh.com):** Architected an intelligent major selection system. Integrated AI to analyze national exam results and psychological assessments across multiple stages, delivering highly optimized academic pathways.
- **B2B Multi-Tenant Portals (entekhab.baazmoon.com & hedayat.baazmoon.com):** Developed portals enabling institutes to manage students, conduct in-depth assessments (Zarebin & Dopamine), and identify academic flaws.
- **Data Engineering & Automated Reporting:** Designed robust pipelines to process massive datasets from national exams. Engineered an automated module generating customized PDF report cards detailing admission probabilities.
- **Tech Stack:** *Python, Django/FastAPI, AI/ML Libraries, PostgreSQL, React, Docker, PM2, Nginx.*

Software Engineer (Part-Time)

Farasima

May 2024 – Present

Tehran, Iran

- **High-Throughput Architecture:** Contributed to the backend engineering of an SMS firewall, optimizing the system to process and rule-match 300,000 to 500,000 messages per second with minimal latency.
- **Gateway & Observability:** Designed gRPC-based interfaces routing messages to AI anomaly detection modules. Leveraged ClickHouse, Prometheus, and Grafana for comprehensive monitoring and ultra-fast analytical storage.
- **Tech Stack:** *Python, gRPC, ClickHouse, Prometheus, Grafana, HAProxy.*

Software Engineer

GradientDp!

Mar 2022 – Dec 2022

Tehran, Iran

- **FinTech Simulation System (B.Sc. Thesis Project):** Designed and maintained a robust paper trading engine tailored for the Iranian stock market, enabling users to backtest strategies using simulated real-time market data flows.
- **Tech Stack:** *Python, PostgreSQL, WebSockets, Market Data Integration.*

AI Trainee

Sharif University of Technology (Advanced Tech Development Research)

Aug 2019 – Sept 2020

Tehran, Iran

- Researched Video Swin Transformer models to improve speed-accuracy trade-offs in video recognition tasks.

Software Engineer

Snapp!

Mar 2019 – Aug 2019

Tehran, Iran

- **Distributed Event Dispatching:** Engineered scalable event dispatching services utilizing polling mechanisms and MQTT protocols within a core dispatching team.
- **Infrastructure Management:** Managed the deployment and maintenance of heavy-load, high-traffic services on Kubernetes clusters.
- **Tech Stack:** *Kubernetes, MQTT, Microservices Architecture.*

Notable Projects

- **Video Swin Transformer:** Implemented a machine vision model leveraging spatial-temporal factorization and self-attention mechanisms to achieve optimal speed-accuracy trade-offs in video behavior recognition.
- **Search Engine:** Engineered a document retrieval system utilizing both statistical models (TF-IDF) and advanced word embeddings (Word2Vec) to efficiently process, index, and answer complex user queries.
- **Evolutionary Game AI (Flappy Plane):** Designed a neuroevolutionary algorithm capable of learning and mastering gameplay across 3 difficulty modes. Implemented custom fitness functions and genetic mechanics to ensure multi-generational survival.
- **Handwritten Digit Recognition:** Built a fully connected Artificial Neural Network (ANN) completely from scratch using core NumPy math operations, featuring custom forward/backward propagation and gradient descent to classify the MNIST dataset.

- **Super Mario Bros in Java:** Developed a 2D platformer desktop application applying strict Object-Oriented Programming (OOP) principles. Utilized Java multithreading and socket programming for custom game state management, multiplayer mechanics, and real-time social music sharing.
- **Message Broker (Dallal):** Designed a lightweight, robust custom message broker and text messenger utilizing Python's Socket library. Implemented a reliable client-server architecture supporting multi-client concurrent communication.

Teaching Experience

Amirkabir University of Technology

Fall 2020 – Spring 2026

- **Head Lab Instructor & Lecturer:** Microprocessor and Assembly Labs (6 consecutive semesters). Authored the comprehensive official lab manual and fully managed the Assembly module, including curriculum design, exam authoring, and student evaluation.
- **Head Teaching Assistant:** Computer Networks (Dr. M. Sabaei, 3 Semesters).
- **Head Teaching Assistant:** Startup Development (Dr. S. H. Ghanadpour, 2 Semesters).
- **Head Teaching Assistant:** Web Engineering (Dr. S. P. Alvani, 2 Semesters).
- **Head Teaching Assistant:** Data Mining (Dr. M. Amirmazlaghani, Fall 2022) | Operating Systems (Dr. S. A. Javadi, Fall 2021).
- **Teaching Assistant:** Internet of Things, Microprocessor and Assembly, Data Structures and Algorithms, Design Algorithms, Advanced Programming.

Certifications & Relevant Coursework

- **Certifications:** Certified Ethical Hacker (CEH) - Maktabkhooneh.
- **Online Specializations (Coursera):** Deep Learning, Machine Learning, Mathematics for Machine Learning.
- **Core Coursework:** Theory of Machines and Languages, Cloud Computing, Computer Networks, Data Mining, Robotics, Principles of Computational Intelligence, Information Retrieval.

Leadership & Soft Skills

- **Technical Strategy & Architecture:** Bridging business requirements with scalable technical solutions, driving microservices migrations, and establishing CI/CD best practices.
- **Team Leadership & Agile Methodologies:** Leading cross-functional teams, orchestrating Agile/Scrum processes, conducting code reviews, and mentoring junior to mid-level engineers.
- **Communication & Technical Mentorship:** Strong technical presentation skills honed through 6+ semesters of academic teaching. Experienced in translating complex engineering concepts for non-technical stakeholders.

Skills & Qualities

- **Programming Languages:** Python 3, Go, C, C++, JavaScript, Java, PHP, $\text{\LaTeX}$.
- **Tools & DevOps:** Docker, Kubernetes, PM2, Airflow, Nginx, HAProxy, Git, GitLab CI/CD, Linux, Bash.
- **Frameworks & Libraries:** GoFiber, Echo, Django, FastAPI, NumPy, Pandas, React.js, D3.js. (Working knowledge: React Native, Flutter)
- **Databases & Messaging:** PostgreSQL, MongoDB, SQL Server, Elasticsearch, ClickHouse, RabbitMQ, Redis.
- **Languages:** Persian (Native), Turkish (Native), English (Proficient - TOEFL-iBT: 108/120).

References

- Dr. Hamed Farbeh** **farbeh@aut.ac.ir**
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“I live in the Middle East. Nothing is impossible for me.”

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